

Real Time Embedded Software (IE420 / EL434 / EL518)

Course Placement: B.Tech (EVD) / B.Tech (ICT) / M.Tech (ICT with Spl. in VLSI & ES)

Credits: 3-0-2-4

Course format: 3 hours lecture and 2 hours lab every week.

Course content: This course introduces the design and implementation of real-time embedded software systems with strict response-time requirements. It covers embedded hardware, real-time operating systems (RTOS), driver configuration, sensor interfacing, and real-time data streaming to cloud-based platforms. Through lectures and hands-on labs, students develop systems used in applications such as healthcare monitoring, smart infrastructure, IoT, and edge computing.

Text Book

- [1] Computers as Components: Principles of Embedded Computing System Design ~ Wolf, Marilyn (Elsevier, 2012)
- [2] Real ~ Time Systems Design and Analysis: Tools for the Required Materials: Grading: Practitioner. Laplante ~ Phillip A. & Seppo J. Ovaska (John Wiley and Sons, 2011)

References

- [1] Q. Li and C. Yao, Real-Time Concepts for Embedded Systems, 1st ed. San Francisco, CA, USA: CMP Books, 2003.
- [2] J. W. Valvano, Embedded Systems: Introduction to ARM Cortex-M Microcontrollers, Volume 1, 5th ed. Lexington, KY, USA: CreateSpace Independent Publishing, 2018.
- [3] J. Yiu, The Definitive Guide to ARM Cortex-M0 and Cortex-M0+ Processors, 2nd ed. Oxford, UK: Newnes, 2015.
- [4] E. White, Making Embedded Systems: Design Patterns for Great Software, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2011.
- [5] A. Bahga and V. Madiseti, Internet of Things: A Hands-On-Approach, 1st ed. Universities Press, 2015.
- [6] AWS, Microsoft Azure, and Google Cloud documentation
- [7] Confluent, The Ultimate Data Streaming Guide, 2025. [Online]. Available: <https://www.confluent.io/resources/ebook/streaming-use-case-guide/>

Assessment Method

Sl. No.	Component	Weightage
1.	Second In-Sem Exam	30%
2.	End-Sem Exam	30%
2.	Assignments & Labs	20%
4.	Project	20%

Course Outcomes (COs)

- CO1: Understand fundamental concepts and distinctions of real-time and embedded systems.
- CO2: Analyze and configure hardware drivers and port real-time operating systems.
- CO3: Apply software abstraction layers and implement real-time dashboards using GUI frameworks.
- CO4: Integrate embedded systems with IoT/cloud platforms and stream sensor data in real-time.
- CO5: Design and develop real-time data streaming system for practical domains.

Program Outcomes (POs) Mapping

Course Outcome (COs)	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1: Fundamentals of Real-Time and Embedded Systems	✓	✓					
CO2: Microcontroller and Driver Setup	✓	✓	✓		✓		
CO3: Real-Time Operating Systems (RTOS)	✓	✓	✓	✓	✓		
CO4: Real-Time GUI and Dashboard Development	✓	✓	✓	✓	✓	✓	
CO5: Cloud Integration for Real-Time Data Streaming	✓	✓	✓	✓	✓	✓	✓

Course Outline

Module	Topic		Hours
1	Fundamentals of Real-Time and Embedded Systems		4
	1.1	Real-Time and Embedded Systems Overview	2
	1.2	Key Concepts, Trends, and Applications in Real-Time Computing	2
2	Microcontroller and Driver Setup		9
	2.1	Microcontroller Architecture and Driver Configuration	3
	2.2	Basics of Low-Level Embedded Programming	2
	2.3	Peripheral Interface Programming	2
	2.4	CI/CD for Embedded Software	2
3	Real-Time Operating Systems (RTOS)		14
	3.1	RTOS Concepts and Architecture	1
	3.2	Building Multitasking Applications Using RTOS	1
	3.3	Task Management in RTOS	1
	3.4	Resource Management in RTOS	1
	3.5	Task Synchronization Techniques	1

	3.6	Event Handling Mechanisms	1
	3.7	CPU Utilization Analysis in RTOS	1
	3.8	Scheduling Algorithms Implementation and Analysis	1
	3.9	Interrupt Management in RTOS	1
	3.10	Time Management Features	1
	3.11	Rate Monotonic Scheduling	1
	3.12	First-Come-First-Served Scheduling	1
	3.13	Round-Robin Schedulers	1
	3.14	Weighted-Round-Robin Schedulers	1
4	Real-Time GUI and Dashboard Development		10
	4.1	Communication Between Firmware and Python	2
	4.2	Scripting Real-Time Device Connections	1
	4.3	GUI Development using Python, Qt, and Kivy	3
	4.4	Dashboard Design and Development for Embedded Systems	1
	4.5	Real-Time Firmware-Dashboard Integration	1
	4.6	Integrating AI/ML with Embedded Dashboards	2
5	Cloud Integration for Real-Time Data Streaming		10
	5.1	Cloud Services for Real-Time Communication	2
	5.2	Using AWS API Gateway, Lambda, and DynamoDB for Automation	2
	5.3	Connecting Devices to Cloud via Wi-Fi	2
	5.4	JSON Structuring, HTTP Methods & Error Handling	2
	5.5	Building End-to-End IoT Solutions with Cloud for Real-Time Data Streaming	2
6	Lab Work and Projects		28
	6.1	Practical RTOS, Sensor Interface, and Dashboard Exercises	16
	6.2	Micro Project Development and Demonstration	12